

Daniel Joseph Williams

Princeton, NJ
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PhD Candidate in Computer Science at Princeton University conducting research at the intersection of artificial intelligence, physics, and clean energy technologies. Focused on building differentiable, GPU-accelerated software for physics-based design.

Education

Anticipated: **Princeton University**
Spring 2029 *PhD Program in Computer Science (Artificial Intelligence / Machine Learning)*
Co-advised by Dr. Ryan P. Adams and Dr. Egemen Kolenen.
Conferred: Master of Arts (MA) in Computer Science (Spring 2026)
May 2024 **University of Maryland, Baltimore County (UMBC)**
Bachelor of Science (BS) in Computer Science
(AI/ML and Data Science Tracks)

Research Experience

Aug. '24 - **Princeton University**
Present *Princeton, NJ*
Computational Mathematics / High Dimensional Geometry / GPU-Accelerated Systems

- Developing GPU-accelerated tools for differentiable geometric computing using JAX, with applications in ML and physics-based design.
- Designing type systems for aggressive GPU kernel optimization with Princeton's PL group.

June '24 - **Oak Ridge National Laboratory (ORNL)**
August '24 *Oak Ridge, TN*
Large Language Models / Knowledge Systems / Energy Infrastructure

- Developed LLM-based question-answering systems to support hydropower facility maintenance.
- Built automated data pipelines for document ingestion, embedding generation, and structured retrieval over facility maintenance records.

June '23 - **Princeton Plasma Physics Laboratory (PPPL)**
Dec. '23 *Princeton, NJ*
High Performance Computing (HPC) / Plasma Physics / Nuclear Fusion

- Built an automated pipeline integrating QSC, DESC, BNORM, and REGCOIL to evaluate millions of near-axis stellarator configurations for the MUSE experiment.
- Investigated ML algorithms for intelligent search over the quasi-symmetric stellarator design space.

June '22 - **Massachusetts Institute of Technology (MIT)**
April '23 *Cambridge, MA*
Artificial Intelligence / Deep Learning / Nuclear Fission

- Developed CNN surrogate models for nuclear reactor core loading optimization, achieving $R^2=0.986$ and a 156-hour speedup over full physics simulation.
- Collaborated with MIT NSE researchers to apply deep learning to reactor safety constraints.

April '21 - **Johns Hopkins Applied Physics Laboratory (JHU APL)**

May '22 *Laurel, MD*

AI Security / Self-Driving Cars / Infrastructure Vulnerabilities

- Developed open source data gathering & analysis pipeline assessing machine learning vulnerabilities in containerized infrastructure for self-driving cars.
- Published research on adversarial attacks and AI-enabled software stack vulnerabilities.

April '21 - **UMBC**

May '21 *Catonsville, MD*

Cybersecurity / Reinforcement Learning / Malware Analysis

- Implemented deep reinforcement learning algorithms for malware detection strategies evaluated in custom OpenAI Gym environments.
- Studied proximal policy optimization (PPO) and Q-learning for cybersecurity.

Publications

- 2025 X. Chu, T. Qian, B. Berlinger, K. Hammond, M. Haque, S. Salia, **D. J. Williams**, E. Zhu, and M. C. Zarnstorff, "Error field measurements and correction on MUSE permanent magnet stellarator," *Plasma Physics and Controlled Fusion*, 2025, doi: 10.1088/1361-6587/ae1870.
- 2022 **D. Williams**, C. Clark, R. McGahan, B. Potteiger, D. Cohen, and P. Musau, "Discovery of AI/ML Supply Chain Vulnerabilities within Automotive Cyber-Physical Systems," 2022 IEEE International Conference on Assured Autonomy (ICAA), Fajardo, PR, USA, 2022, pp. 93-96, doi: 10.1109/ICAA52185.2022.00020.
- 2022 **D. Williams**, C. Clark, R. McGahan, B. Potteiger, L. Gonzalez, P. Musau, N. Potteiger, & R. Cohen, "Reachability-Based Monitoring for Protection against AI/Machine Learning Supply Chain Vulnerabilities within Cyber-Physical Systems," *Decoded – JHU APL Reverse Engineering Magazine*, Issue 7, pp. 45-52, February 2022.

Selected Presentations

- 2023 "Optimizing Permanent Magnet Stellarators with Machine Learning," American Physics Society Division of Plasma Physics (APS DPP) Meeting. *Accepted abstract and poster available upon request.*
- 2023 "A Novel Framework for Accelerating Core Loading Pattern Optimization with Deep Learning Surrogate Models," American Nuclear Society (ANS) Student Conference (Award Winning Research Talk). *Accepted 4-page summary available upon request.*
- 2022 "AI/ML Supply Chain Vulnerabilities in Automotive Cyber-Physical Systems," IEEE ICAA (30 min. Technical Presentation). *Refer to the Publications section for conference details.*

Honors/Awards

- Francis Robbins Upton Fellowship - 2024
- GEM Graduate Fellowship - 2024
- NASA Maryland Space Grant Recipient - 2024
- 1st Place – Technical Research Exhibition at NSBE Fall Regional Conference (2023)
- 1st Place - Best Undergraduate Lightning Talk in AI at ANS Student Division (2023)
- UMBC Meyerhoff Scholar (M32)

Technical Skills

Languages Python, C/C++, Java, JavaScript (Node.js), Swift

Frameworks JAX, PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn
Systems GPU Programming, High-Performance Computing (HPC), Linux Administration, Docker, AWS, Xcode
Hardware NVIDIA Jetson, Raspberry Pi, Arduino
Tools Jupyter, LaTeX, SLURM, Git/GitHub

Leadership

Leadership Princeton Black Graduate Caucus Secretary 2025-2026, UMBC National Society of Black Engineers Vice President 2023-2024, UMBC Cyber Defense Club President 2022-2023

Teaching Experience

- Princeton COS 126 - Intro Computer Science Class Teaching Assistant (TA) (2025-2026)
- Princeton AI4ALL Summer Course - Guest Lecturer (2025)